

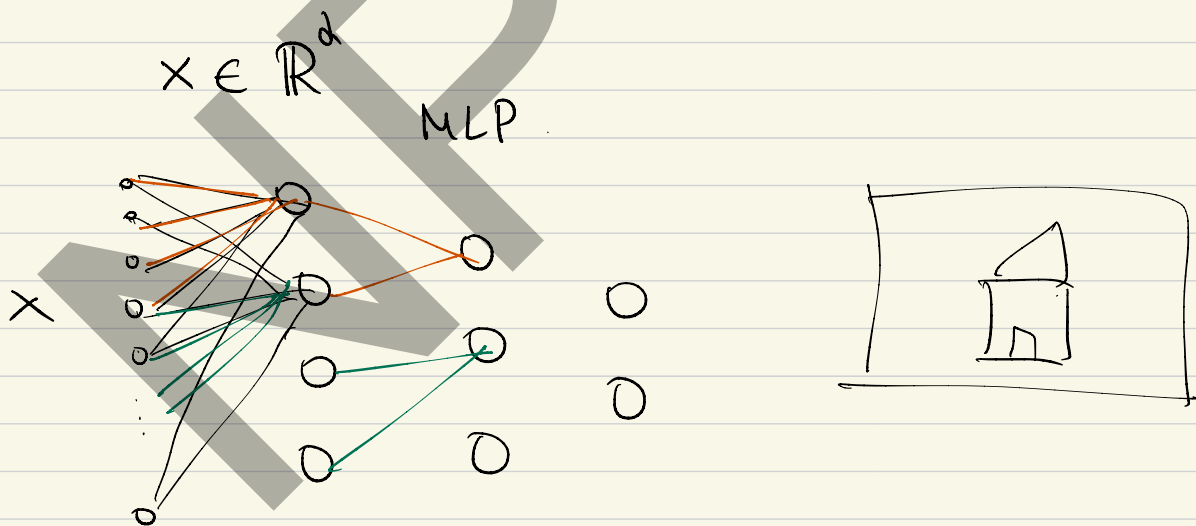
Convolutional Neural Networks

Regularized MLP suited for "grid-like" topological data.

Eg: Image, 3D volumes

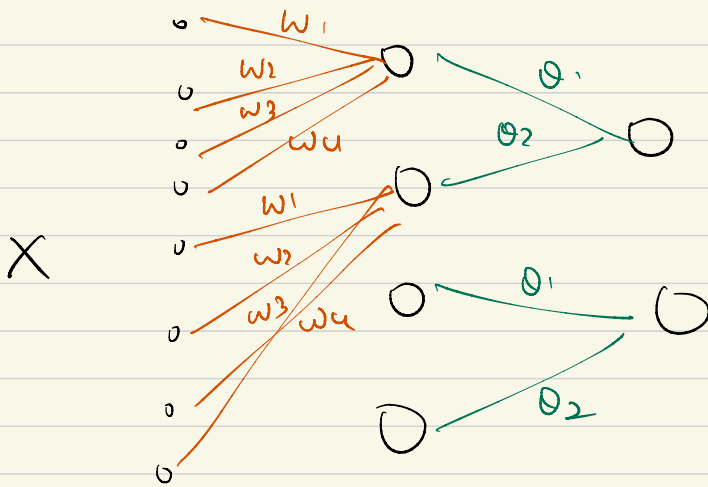
CNNs: MLP with two specific "hard" regularizers on the parameter space

- i) Local Receptive field
- ii) Parameter sharing

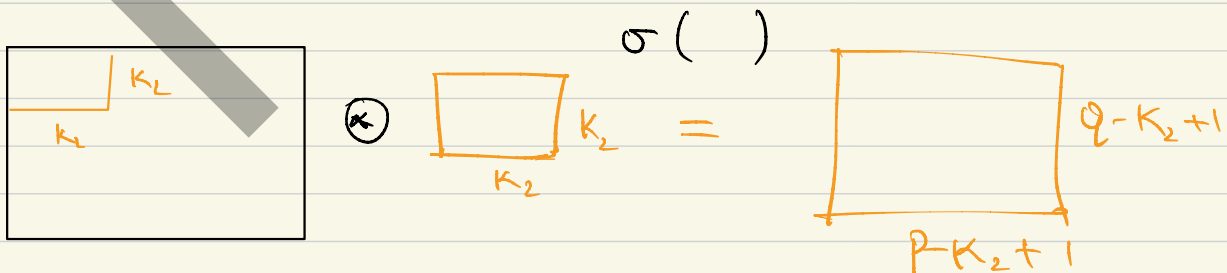
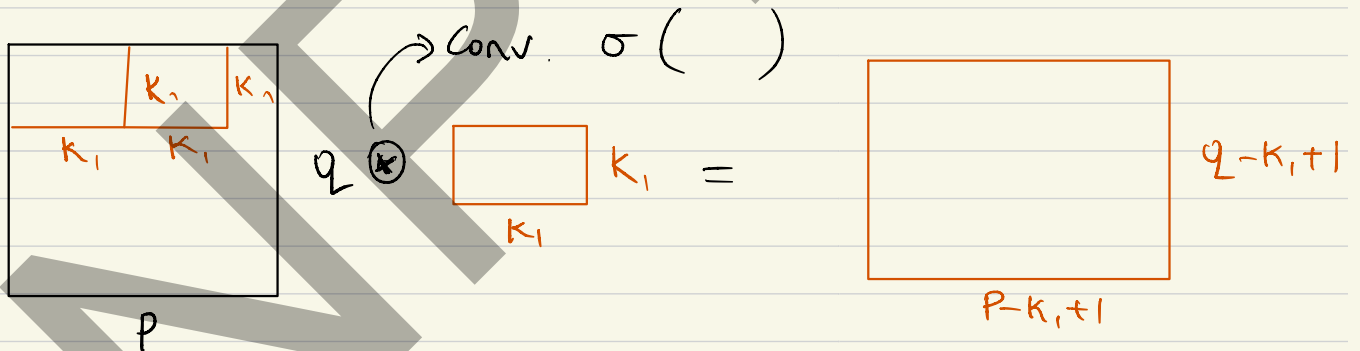


Local Receptive field: Connect only a subset of data/activations to every neuron in each layer.

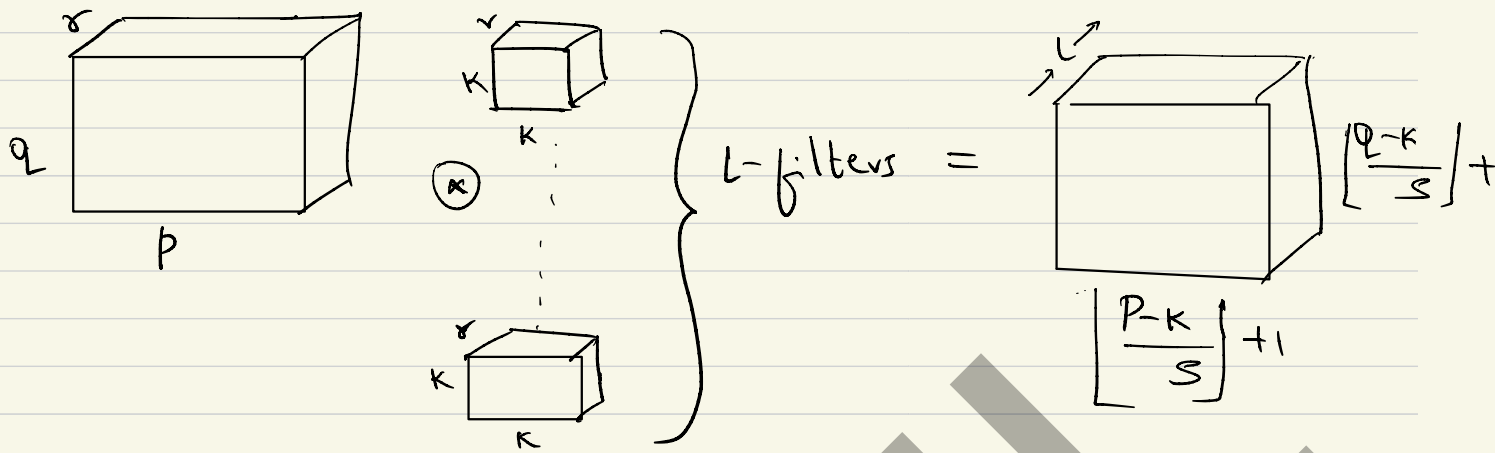
ii) parameter-sharing



Parameters of every receptive field is kept the same.



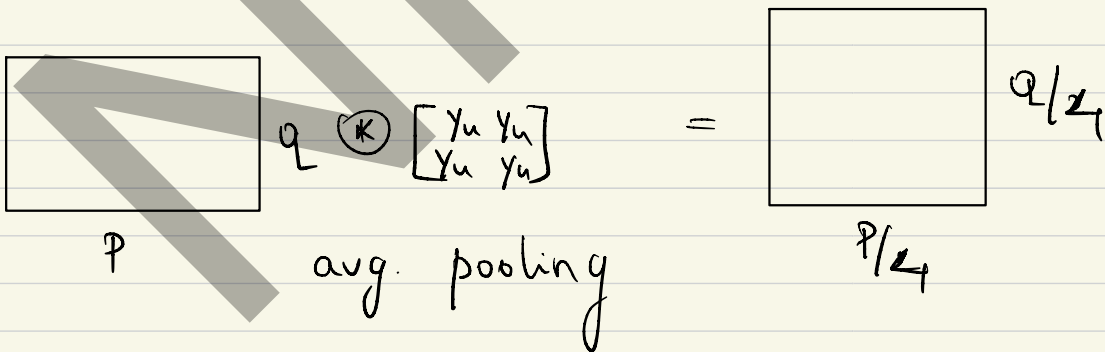
CNN



hyper params :

- no of filters in each layer
- filter size
- stride / padding
- no of layers / depth

pooling operations.



avg. pooling

max. pooling

Examples of CNN- End- to- End.

i) Classification



ERM.

LeNet, Alexnet, Resnet, Inceptionnet

ii) Fully Convolutional.

Semantic - segmentation.

